

SWAYAM MEHTA

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Backend Software Engineer | Distributed Systems & Full-Stack Infrastructure

EDUCATION

M.S. Computer Science	2024 - 2026
Arizona State University	4.0/4.0 GPA
B.Tech. Computer Science and Engineering	2020 - 2024
Vellore Institute of Technology	8.65/10.0 GPA

SKILLS

Programming	Python, Java, JavaScript, TypeScript, SQL, Kotlin, C++
Backend	FastAPI, Flask, Celery, REST APIs, PostgreSQL, Redis
Frontend	React, Next.js, D3.js
Infra/Tools	Docker, Kubernetes, AWS, Linux, Git, Prometheus, Grafana

EXPERIENCE

JobTalk AI: Machine Learning Engineering Intern	03/2026 - 05/2026
- Built an async resume ingestion service (FastAPI, Celery, S3) handling multi-format uploads with tenant-isolated storage and PII-at-rest encryption for a multi-client SaaS deployment. - Diagnosed a Celery/asynpg connection-pool exhaustion bug causing failures under load; fixed it and lifted end-to-end pipeline success to 99%+. - Deployed Prometheus + Grafana monitoring across the pipeline, tracking throughput, latency percentiles, and queue depth in real time.	
Vellore Institute of Technology: Research Assistant	01/2024 - 05/2024
Built a stock price forecasting pipeline (Bi-LSTM, GRU, GAN) and served forecasts and sentiment scores through a Flask REST API, reducing RMSE 40% over prior baselines.	
Hewlett Packard Enterprise: CTY Intern - High Performance Computing	01/2023 - 06/2023
- Built a Kubernetes runtime security system with Falco/eBPF to detect container escapes, privilege escalation, and filesystem tampering, streaming alerts to Prometheus/Grafana with Slack/Discord notifications. - Cut incident detection latency ~40% and increased detection coverage ~30% by replacing manual log review with automated behavioral alerts.	

PROJECTS

ClickLess AI [Python, LangGraph, Mistral, Neo4j, Playwright, Browser Use, Streamlit]	Spring 2026
- Designed a multi-agent pipeline that turns a natural-language grocery request into a submitted cart: NLU parsing → retailer API lookup → knowledge-graph ranking → automated checkout via Playwright. - Built an encrypted credential vault issuing short-lived session tokens at checkout instead of storing retailer logins.	
Quizzi [Next.js, TypeScript, PostgreSQL, Supabase, Redis, D3.js]	Summer 2025
- Redesigned PostgreSQL row-level security policies to remove per-row re-evaluation, cutting policy evaluation from O(N) to O(1) and speeding up large-dataset queries 10–100x. - Cut daily database egress 76% by replacing heavy JOINS with lightweight key-based queries, client-side assembly, and TTL caching.	
Donna [FastAPI, LangGraph, Twilio, LiveKit, Groq, Cartesia]	Fall 2025
- Implemented a real-time AI phone assistant with sub-second end-to-end voice response latency, integrating Twilio SIP trunking, LiveKit WebRTC audio transport, a LangGraph agent pipeline with Groq inference and Cartesia TTS. - Containerized and deployed the voice stack on a resource-constrained VM under production firewall restrictions, implementing failure-safe recovery and graceful session teardown to maintain call continuity.	

LEADERSHIP

IEEE Computer Society - VIT: Secretary	01/2023 - 12/2023
Led the team that launched Chota Dhobi, a laundry-tracking app reaching 24K+ active users and 15% month-over-month growth.	